
UNI HUB DIGITAL PLATFORM – PROJECT DOCUMENTATION

1. Introduction

Project Overview

UniHub is a web-based digital platform designed to connect university students within a secure and verified environment. The platform allows users to:

- Join a university-specific hub using their institutional email
- Participate in events
- Buy and sell items within their campus community

The system enforces strict domain-based authentication to ensure that only verified users access their respective university hubs.

Objectives

- Provide a **secure digital environment** for university communities
- Enable **easy buying and selling**
- Support **event management, opportunities and participation**
- Ensure **fast performance and ease of use**
- Allow new users to **learn and use the system within 3 minutes**

Scope of the System

The system covers:

- User authentication and role management
- Event creation and participation
- Marketplace (buying and selling within same university)
- Admin controls and monitoring

2. System Requirements

Functional Requirements

- System shall users to create accounts using their name, email and password
- User shall type a keyword and see all matching results
- User can add items, change quantities and remove items
- System calculates total and processes mobile payments

2.2 Non-Functional Requirements

Usability

- The system shall be simple **and intuitive**
- A new user should be able to:
 - Sign up
 - Verify email
 - Access dashboard
 - Browse listings/events**within 3 minutes**
- Clear navigation menus and buttons
- Minimal clicks (max 3–4 actions to complete a task)

Performance

- System must load **within 8 seconds**
- Pages should load in **less than 3 seconds after first load**
- Optimized images and database queries

Security

- Email domain validation (e.g.,@std.must.ac.ug)
- Password encryption using hashing
- Email verification via OTP (One-Time Password)
- User-specific permissions (seller-only edits, admin privileges)
- Special **listing security key** for sellers

Reliability

- System available 24/7
- Backup of database
- Error handling for failed operations

Scalability

- Ability to add multiple universities (multi-hub system)
- Efficient database design for future growth

3. System Architecture

Technology Stack

- **Frontend:** HTML, CSS, JavaScript
- **Backend:** PHP
- **Database:** MySQL
- **Server:** WAMP/XAMPP

Architecture Design

- Client (Browser) → Frontend (HTML/CSS/JS)
- Server (PHP) handles logic
- Database (MySQL) stores data

System Flow

1. User signs up
2. Email is verified
3. Domain is checked
4. User is assigned to university hub
5. Access granted to:
 - Events
 - Marketplace

4. System Modules

User Management and Authentication

- Email-based signup
- Domain validation
- OTP email verification
- Role-based access (Member/Admin)

Key Feature:

Only users with valid university domains can access the hub.

Events Module

- Admin creates events
 - Users view upcoming events
 - RSVP functionality
 - Event management (edit/delete by admin)
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4.3 Marketplace (BS) Module

- Users create listings (title, price, category, images)
- Listings visible only within same university
- Seller controls:
 - Edit
 - Delete
 - Mark as sold

Security Enhancement:

- Each seller has a **unique listing key**
 - Prevents unauthorized edits
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Messaging System

- Buyers and sellers communicate per listing

- Conversation stored in database
 - Private and secure messaging
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Admin Module

- Dashboard with:
 - Total users
 - Events
 - Listings
 - Manage users (edit/delete roles)
 - Delete inappropriate listings
 - Add new university domains
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5. Database Design (Simplified)

Main Tables

- Users (id, email, password, role, university_id)
 - Universities (id, name, domain)
 - Events (id, title, date, location)
 - Listings (id, title, price, seller_id, status)
 - Messages (id, listing_id, sender_id, message)
 - RSVPs (user_id, event_id)
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6. User Interface Design

Design Principles

- Clean layout
 - Easy navigation
 - Mobile responsive
 - Minimal clutter
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Main Pages

- Home Page
- Sign Up / Login Page

- Dashboard
 - Events Page
 - Marketplace Page
 - Admin Panel
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User Experience Goal

A new user should:

1. Register
 2. Verify email
 3. Log in
 4. View marketplace and current event, all in less than two minutes
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7. Performance Optimization

- Use compressed images
 - Limit image uploads (max 3 per listing)
 - Optimize SQL queries
 - Use caching where possible
 - Minify CSS and JavaScript
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8. Security Features

- Password hashing (bcrypt)
 - Session management
 - Email verification
 - Domain restriction
 - Role-based authorization
 - Seller-specific permissions
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9. Testing Strategy

Types of Testing

- Unit Testing (PHP functions)
- Integration Testing

- User Acceptance Testing (UAT)

Sample Test Cases

- Invalid email domain → Access denied
- Unverified email → Cannot login
- Non-seller editing listing → Blocked
- RSVP toggle → Works correctly

10. Deployment

Tools

- WAMP Server
- phpMyAdmin
- Web browser

11. Future Improvements

- Mobile App version
- Payment integration (credit card)
- Notification system (SMS/Email)
- AI-based recommendations

12. Conclusion

UniHub provides a secure, fast, and user-friendly platform for university students to interact digitally. By focusing on:

- **Speed (under 8 seconds load)**
- **Ease of use (under 3 minutes learning)**
- **Security (domain + verification)**

the system ensures a reliable and scalable solution for campus digital engagement.